Physical Chemistry Cume

October 1, 2010

The wave function of a 1s-electron in a Hydrogen atom is $\Psi = N\exp(-r/a_0)$, where N is a constant, $a_0 = 52.9$ pm, and the variable r is the distance from the nucleus.

For the electron in the Hydrogen atom 1s-state:

a) Normalize Ψ.

b) Determine at which point in space it is most likely to find the electron.

c) Calculate the probability of finding the electron inside a small volume of magnitude 1.0 pm^3 located at a distance a_0 from the nucleus.

d) Calculate the average value of the distance of an electron from the nucleus.

Each question 25 points

Help:

$$\int_0^\infty x^n e^{-ax}\, dx = n!/a^{n+1}$$